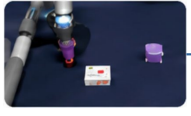
(a) Controlled simulator replays

Illustrative rendered views



Shared recorded qpos / qvel

a0	a1	...	aT-1

Same sequence of raw actions

N: nominal



M: mass and inertia x 4



D: delay of 2 raw steps



Paired replays reuse recorded configurations and actions across N, M, and D.

(b) Fixed-budget training

S: State coverage

192 source records; N only
384 window records

P: Paired physics

64 sources; each under N / M / D
128 + 128 + 128 window records

U: Unpaired physics

192 records; one condition each
128 + 128 + 128 window records

Same model and optimizer

10,000 updates; 3 seeds per policy

(c) Strict evaluation

64 held-out source episodes

Zero training-episode overlap

N | **M** | **D** | **MD**

MD: held-out composition of familiar settings

Latent prediction

Raw / standardized MSE; cosine error
Teacher-forced and free-running

Physical position evaluation

Common train-only linear decoders
Block / end effector; persistence

18 forecast steps = 90 raw action steps